An On-Chip Reconfigurable Front-End for Ultra-Low-Power RF Energy Harvesting

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***Abstract*—This brief presents a reconfigurable dual-output front-end for ultra-low-power (ULP) RF energy harvesting applications, demonstrating a stable output voltage ranging from 1.3 V to 1.8 V over a wide input power range of –22.5 dBm to –3 dBm. The proposed design incorporates a novel knockout-based reconfigurable architecture with dual outputs and a self-compensation dual-stage cross-coupled differential-drive (CCDD) rectifier topology, enabling efficient output control for ultra-low-power systems while maintaining high efficiency and sensitivity. Implemented in 40 nm CMOS technology, the 4-stage 900 MHz rectifier achieves a peak power conversion efficiency (PCE) of 42.8% at –20 dBm input with 0.5 MΩ load, along with an measured –23.6 dBm sensitivity at 1 V output voltage under 1 MΩ load.**

***Index Terms*—RF energy harvesting (RFEH), ultra-low-power (ULP) devices, wide stable power range, CCDD rectifier.**

I. Introduction

THE RAPID advancement of Internet of Things (IoT) and medical devices has revolutionized modern society with unprecedented convenience, while inadvertently creating billions of sensor nodes and imposing significant energy burdens. Fortunately, recent advancements in ultra-low-power (ULP) sensors and high-efficiency energy harvesting technologies have emerged to extend operational lifetimes of these devices. Among these solutions, RF energy harvesting (RFEH) has emerged as a feasible power solution for self-powered systems, owing to its environmental independence and ubiquitous availability in urban environments.

However, as depicted in Fig. 1, ambient RF power density is constrained. When accounting for impedance matching and rectification losses, the available power for core circuits can only sustain microwatt-level, demanding ultra-low quiescent currents (I_Q). To address this, some works [1] have adopted event-driven and power-gating techniques to minimize active duty cycles and reduce average power consumption. Nevertheless, power management unit (PMU) must maintain continuous operation for emergency responsiveness. While prior works [2], [3], [4] have proposed various solutions achieving nW-to-pW-level quiescent power and fast transient response speed, most of them rely on CMOS transistors working in subthreshold region, which leads to significant process variation sensitivity and limited stability within narrow supply voltage range. Consequently, these ULP-PMUs prove difficult to be effectively driven by conventional rectifiers, as rectifiers' output voltages exhibit significant variations corresponding to changes of input RF power (P_{RF}).

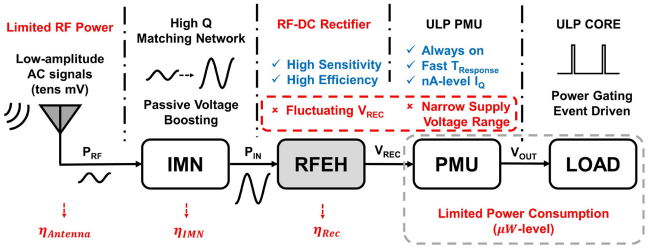

Fig. 1. Proposed ULP-RFEH system.

To tackle energy harvesting challenges in low-input conditions and complex RF environments, various circuit-level solutions have employed threshold voltage (V_{TH}) compensation techniques. Authors in [5], [6], [7], [8], [9] optimized the V_{TH} by choosing low-V_{TH} transistors and adjusting the gate or substrate voltage to enhance charging currents or mitigate leakage, thereby improving power conversion efficiency (PCE) and sensitivity of the rectifier. At the system level, various reconfigurable architectures are commonly used in [10], [11], [12], [13]. While these approaches effectively extend the input power range for high PCE, they also transfer the instability of the rectifier's output voltage to the subsequent PMU, requiring complex boost/buck converters that introduce additional power losses and reduce efficiency of the energy loop. The authors in [14] proposed a tri-band RFEH system with a voltage limiter to achieve controlled output ranges; however, this results in significant efficiency degradation at higher input power levels.

In this brief, we present an 4-stage reconfigurable dual-output CCDD RFEH front-end that incorporates both static and dynamic compensation to achieve high sensitivity and high PCE at low RF input range. The proposed design utilizes a low-complexity controller to reconfigure the rectifier into main and secondary paths, ensuring a stable main output voltage over a wide input power range while minimizing power loss at higher input levels. Section II explains the operating

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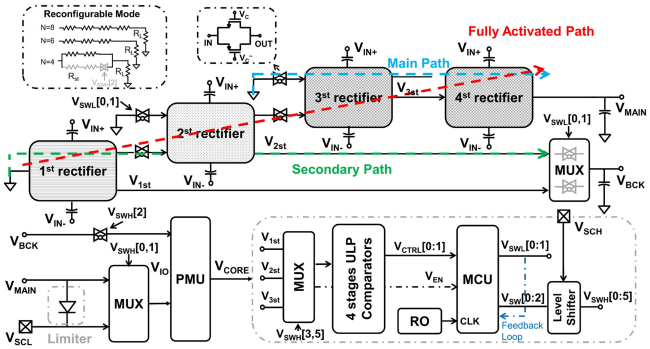

Fig. 2. Architecture of the proposed reconfigurable dual-output rectifier.

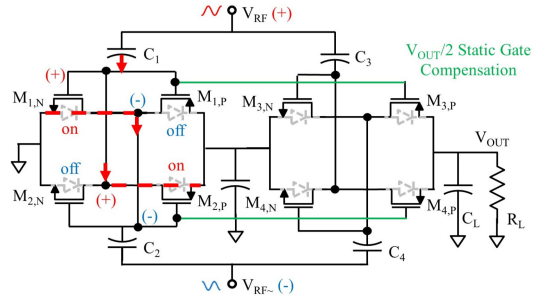

Fig. 3. Self-compensation dual-stage CC rectifier.

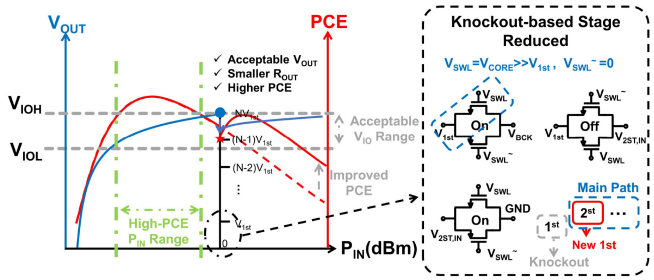

Fig. 4. Principle of knockout-based on-chip reconfiguration.

principle and optimization of the proposed rectifier, followed by measurement results in Section III. Finally, Section IV concludes the work.

II. Implementation of the Proposed Rectifier

Fig. 2 illustrates the proposed dual-output 4-stage rectifier design. Compared to other reconfigurable rectifiers [12], [13], the proposed rectifier enables on-chip implementation of three reconfiguration states (2-3-4 stages), thereby eliminating the need for complex boost/buck DC-DC converters while effectively regulating the main output voltage within a range suitable for ULP-PMUs. This design significantly reduces overall system power consumption while enhancing long-term stability under dynamical energy conditions.

Furthermore, the RFEH front-end employs a novel reconfiguration solution to minimize dependence on extra high-voltage control signals. To enhance energy storage and management, two supercapacitors with different output voltages are incorporated. The high-voltage supercapacitor (V_{SCH}) powers the level shifter, which selects the PMU supply voltage. Additionally, in the 2-stage reconfiguration mode, the main and secondary storage capacitors can be connected in parallel through a switch, maintaining a stable supply voltage while addressing the transient high-current demands of low-power systems.

A. Principle of Proposed Rectifier

The topology of each rectifier stage in the proposed RFEH system is shown in Fig. 3. A CCDD structure is employed to achieve high PCE under low RF input conditions [5]. To overcome the limitation of transistor threshold voltage (V_{TH}), native N-channel transistors ($M_{\mathrm{i,N}}$) are used to provide a close to zero-V_{TH}, while low-V_{TH} (LVT) P-channel transistors ($M_{\mathrm{i,P}}$) are selected. However, due to the different types of N-type and P-type transistors selected, the V_{TH} of the LVT PMOS transistors may be 100 to 200 mV higher than that of the native NMOS transistors for the same gate length. Therefore, in addition to transistor gate-length optimization for achieving optimal threshold voltage, the rectifier also connects the gates of $M_{3,\mathrm{P}}$ and $M_{4,\mathrm{P}}$ to the gates of $M_{1,\mathrm{P}}$, and $M_{2,\mathrm{P}}$, respectively, to implement an additional static compensation of $V_{\mathrm{OUT}}/2$, thereby enhancing the rectifier's performance.

The PCE of a rectifier is shown in equation (1):

$$PCE = \frac{P_{ANT} - (P_{COND} + P_{LEAK} + P_{DYN})}{P_{ANT}} \tag{1}$$

where conduction losses (P_{COND}) are caused by the forward voltage drop and the resistance of the switch, leakage losses (P_{LEAK}) are associated with the reverse leakage current and switch leakage, and dynamic losses (P_{DYN}) are the intrinsic losses during the charging and discharging process [5]. Although static compensation can reduce conduction losses and switches can provide more effective control, they both introduce additional losses that require a trade-off.

B. Principle of Knockout-Based On-Chip Reconfiguration

Fig. 4 illustrates the principle of on-chip reconfiguration and stable output. Due to the energy-efficient design of the low-power system, which reduces the supply voltage, the amplitude of the switch control signals and the power supplied to the core circuitry are also reduced. Despite employing CMOS transmission gates for switching, it is challenging to directly compare and shut off the high output of the rectifier. Therefore, a "knock-out" reconfiguration mode is adopted, where only the first stage of the main rectifier path is monitored. In this reconfiguration mode, due to the limited output voltage of each stage, the output voltage amplitude at switch-connected nodes is constrained to maintain a minimum difference of $V_{TH(SW)}$ relative to the on-chip V_{Power} signal, ensuring sufficient switching on and off to minimize the conduction losses associated with switch resistance.

When the output voltage of a single stage, V_{out}/N, exceeds the preset comparison voltage, the output of the current first stage is disconnected from the input of the second stage and connected to the secondary path, V_{BCK}. The input of the current second stage is then grounded, and after stabilization,

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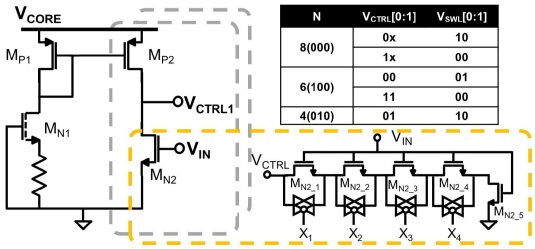

N	$V_{CTRL}[0:1]$	$V_{SWL}[0:1]$
8(000)	0x	10
	1x	00
6(100)	00	01
	11	00
4(010)	01	10

Fig. 5. ULP comparators (left), control logic (top-right) and model correction by adjusting the length of M_{N2} (bottom-right).

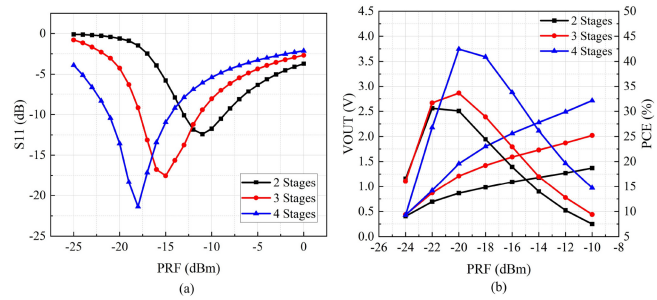

Fig. 6. Simulated results for different stages rectifier versus input RF power with 0.5MΩ resistor load: (a) Return loss (S_{11}), (b) Output voltage (V_{OUT}) and power conversion efficiency (PCE).

it becomes the new first stage of the main rectifier path. Through this reconfiguration mode, as shown on the left side of the figure, the main rectifier output is constrained between the preset V_{IOH} and V_{IOL}. This ensures stable supply voltage for the subsequent ULP-PMU. At the same time, reducing the number of stages lowers the rectifier's output resistance, compensating for efficiency losses, and thereby achieving a high PCE over a wider input power range. Notably, despite the output voltage variations induced by knock-out operations, the rectifier's small inter-stage capacitances enable the first-stage output to stabilize rapidly after knockout, thus ensuring reliable comparator decision-making.

C. Circuit Realization

Fig. 5 illustrates the ULP comparators and their associated logic implementation for rectifier stage control. The nA-level bias current is generated by zero-biased native transistor M_{N1}, while M_{N2}'s 4-bit digitally-tuned dimensions (x<1:4>) generate process-compensated reference voltages for accurate comparison. The output voltages of each rectifier stage, V_{1st}, V_{2st}, are selected through a MUX to ensure that the output voltage of the current first stage in the main path is compared against the predefined reference values. When the first-stage output of the main path exceeds the first predefined reference, Comparator 1 outputs a logic low (0), prompting the MCU to generate the switching control signals that disables the current first stage, thereby reducing the rectifier stages to $N-1$. Similarly, when the first-stage output voltage keep increasing, both Comparators 1 and 2 output a logic low, further decreasing the rectifier stage count to $N-2$. In this scenario, the two disabled stages are reconfigured into a new two-stage rectifier. The control logic and equivalent resistance representations for different stage configurations are depicted in Fig. 5 (right). It is worth noting that the ULP-PMU and subsequent circuit modules operate within a constrained supply voltage range (V_{IO}), spanning from V_{IOL} to V_{IOH}. As a result, the comparison and reconfiguration window is relatively narrow. In certain cases, this limitation may induce stage oscillation, wherein $V_{MAIN}(N) > V_{IOH}$ while $V_{MAIN}(N-1) < V_{IOL}$. To address this issue, the reference voltages for the comparators are carefully pre-defined, while the control signals are fed back to the MCU to ensure system stability and prevent oscillatory behavior. As the input RF signal continues to increase, the output voltage amplitude of a single stage gradually approaches the amplitude of switch control signal (V_{CORE}), rendering the switch ineffective. Therefore, the proposed rectifier supports three basic reconfiguration modes: 2-3-4-stage. Additionally, to prevent circuit failure under higher input power levels, a diode is placed between V_{MAIN} and V_{SCL} for voltage protection. This diode is also charged when excess output energy is available, extending the operation time and enhancing system stability.

Furthermore, an additional design is implemented for the secondary path to optimize energy utilization. When the stage number of secondary path is low, the output of secondary path (V_{BCK}) charges a bypass capacitor to accumulate energy. As the stage count increases to 2, the high V_{BCK} can be connected in parallel with the main path via an auxiliary switch, reducing the rectifier's output impedance and supplying the ULP circuit with the required transient high current. The control signal for this switch is level-shifted to V_{SCH} to ensure proper switching between V_{BCK} and V_{MAIN}. Similarly, this high-voltage switch control signal is also utilized for selecting the supply voltage of the PMU and the input for comparators.

Fig. 6 shows the simulated large-signal S-parameters, output voltage, and PCE of the rectifier with different stage configurations (main path only) under a fixed 0.5 MΩ load. Using a predefined matching network, Fig. 6 (a) depicts the return loss (S_{11}) of the RF front-end. By reconfiguring the rectifier stages, the front-end maintains matching over a wider input power range. As illustrated in Fig. 6 (b), at extremely low input RF power levels, the transistors in the rectifier fail to turn on effectively due to the limitations of V_{TH} and extremely low input voltage amplitude, which leads to additional conduction losses (P_{COND}) when higher stages are used, ultimately reducing both the output voltage and efficiency. Furthermore, since this brief adopts a CCDD architecture to achieve high sensitivity and PCE under low input conditions, the efficiency at high input power levels significantly decreases. Although reconfiguration enables the system to maintain good S_{11} and provide an acceptable supply voltage under various input conditions, it becomes challenging to sustain high efficiency across a broad input power range.

III. Experimental Results

This brief presents an on-chip reconfigurable rectifier designed and fabricated in a 40 nm CMOS process, occupying a core area of $185 \times 200\,\mu m^2$. The design integrates 2 comparators, an on-chip controller and a switch array. Fig. 7 (a) illustrates the chip measurement platform. A signal

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TABLE I
COMPARISON OF PROPOSED RECTIFIER WITH STATE-OF-THE-ART WORKS

	JSSC'17 [10]	JSSC'19 [13]	TCAS-II'20 [12]	TVLSI'22 [15]	TCAS-II'23 [16]	This Work
CMOS Technology	180 nm	180 nm	130 nm	130 nm	180 nm BCDMOS	40 nm
Frequency	915 MHz	915 MHz	820 MHz	900 MHz	915 MHz	900 MHz
Rectifier Topology	Dickson	Dickson	Dickson	Dickson	CCDD	CCDD
Number of Maximum Stages, N	8	12	8+Auxiliary 2	12	4	4
Number of Reconfiguration, N_R	4 (1-2-4-8)	5 (2-3-4-6-12)	4 (1-2-4-8)	2 (6-12)	3 (1-2-4)	4 (2-2 Parallel-3-4)
Extra Power Supply	Yes, 1.8V for switching signals	Yes, 0.5-1.8V for SoC& Loading Control	Yes, for Off-chip MCU	NA	Yes, for digital signal	Yes, for start-up & level shifter
Power Output Usage	Start-up, PMU, load	Load, power storage	Load, Mux controller	Load	Load	Load, PMU, power storage
Stable Out Range (1.3V-1.8V)	NA	-17 to -14.5 dBm @ 1 MΩ	NA	-20 to -18.5 dBm @ 1 MΩ	1 to 5 dBm @ 1 MΩ	-22.5 dBm to -3 dBm @ 1 MΩ
Sensitivity	-14.8 dBm @ 1V (1 MΩ)	-17.8 dBm @ 1V (1 MΩ)	-22 dBm @ 0.3V (0.22 MΩ)	-21.7 dBm @ 1V (1 MΩ)	-10 dBm @ 1V (Capacitor load)	-23.6 dBm @ 1V (1 MΩ)
Maximum PCE	14,18,25,25% (@ -11, -8, -5, 0 dBm)	34.4% @1.3 dBm	39% @-5 dBm	56.7% @2 dBm	34.93% @-10 dBm	42.8% @-20 dBm

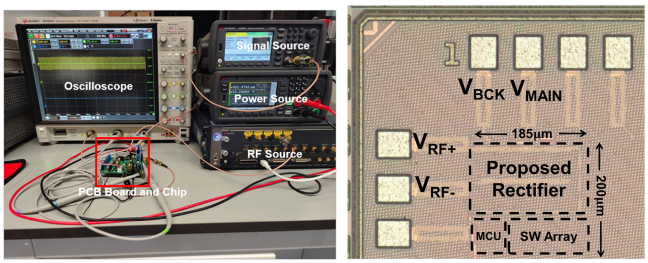

Fig. 7. (a) Measurement setup, (b) Chip photograph of proposed rectifier.

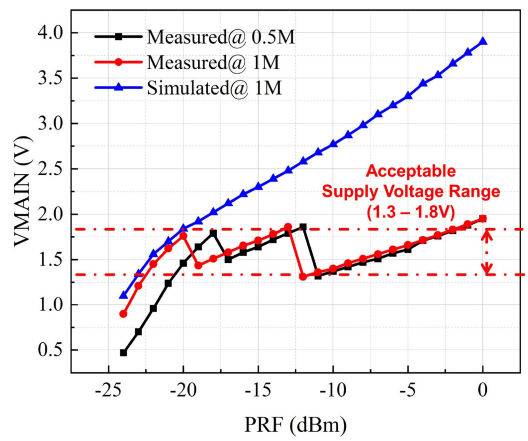

Fig. 8. Measured main output voltage (V_{MAIN}) of proposed reconfigurable rectifier and simulated V_{MAIN} of fully activated rectifier versus input RF power (P_{RF}) for different resister load (RL).

generator is connected to a balun via an SMA connector to provide a fully differential 900 MHz RF signal. After amplification through an off-chip matching network, the signal is fed into the rectifier input. To facilitate testing, the PMU power supply was switched to an external 1.5 V source to ensure continuous activation of the reconfigurable control module. However, in practical applications, this power supply can be switched to the rectifier output voltage V_{MAIN}, enabling complete self-powering and on-chip reconfiguration. To evaluate the PCE, the rectifier output was disconnected from the off-chip storage capacitor. The voltages V_{BCK} and V_{MAIN} were then connected to different resistive loads to measure the actual output power under different configurations. An oscilloscope was used to observe the output voltages of rectifier. Fig. 7 (b) presents a chip micrograph of the proposed reconfigurable rectifier.

Under the condition that the reconfigurable module is activated and the resistance loads are set to 1MΩ and 0.5MΩ, respectively, the measured rectifier's main output voltage V_{MAIN} versus the input RF power (P_{RF}) is shown in Fig. 8. In contrast to the post-layout simulation results of the fully activated 4-stage rectifier, where the output voltage increases monotonically with rising input power, the measured output exhibits a deviation attributed to additional parasitic effects introduced by off-chip components, resulting in an approximate 1 dBm reduction in sensitivity at 1 V output under a 1 MΩ load. Nevertheless, aided by the on-chip MCU and comparators, the rectifier autonomously transitions into 3-stage and subsequently 2-stage reconfiguration modes in response to increasing input power. This dynamic reconfiguration enables the output voltage to remain stably regulated within the predefined target range of 1.3 V to 1.8 V. To regulate the output voltage and prevent control loop oscillations, reference voltages of 0.45 V and 0.65 V are predefined. When the detected 1-stage output exceeds these thresholds, the rectifier transitions into the 3-stage and 2-stage reconfiguration modes, respectively. Notably, each rectifier stage incorporates a dual-stage rectification unit with self-threshold voltage compensation, as illustrated in Fig. 3. Experimental results demonstrate that within an input power range of –22.5 dBm to –3 dBm, the reconfigurable rectifier consistently maintains an output voltage between 1.3 V and 1.8 V. Compared to a conventional fixed 4-stage rectifier,

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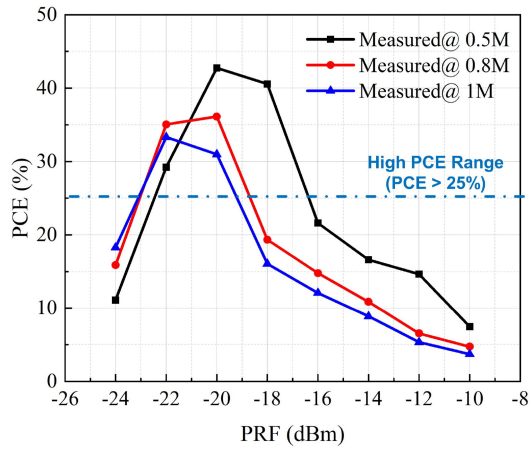

Fig. 9. Measured PCE versus input P_{RF} for different resistor loads.

the proposed design extends the stable power supply range from –20 dBm to –3 dBm, significantly enhancing system robustness across a broader range of RF input power levels.

Fig. 9 presents the efficiency curve of the rectifier as a function of input power. Due to the rectifier's fully active mode operation under low-input conditions with a fixed output impedance, its power conversion efficiency (PCE) demonstrates significant variation across different load resistances. Compared to conventional fixed-stage rectifiers [5], [17], [18], the introduction of numerous switches incurs additional energy losses. Nevertheless, the proposed system achieves an efficiency of 42.8% at an input power of –20 dBm with 0.5 MΩ load. Although the inherent structural limitations of the CCDD architecture constrain the rectifier's efficiency under high RF input conditions, it maintains an efficiency of over 25% within the input power range of –23 dBm to –17 dBm.

Table I compares and summarizes this brief with previous state-of-the-art reconfigurable rectifiers. Our reconfigurable rectifier operates over a wide input power range while maintaining a stable output voltage of 1.3 V to 1.8 V, suitable for ultra-low-power circuits. By leveraging a CCDD structure with dual compensation and utilizing low-threshold voltage PMOS and native NMOS transistors, the rectifier achieves high sensitivity and exceptional efficiency under low-input conditions. Furthermore, through the innovative knockout reconfiguration mode, the rectifier enables complete switch-off at a standard 1.2 V supply, effectively reducing conduction losses introduced by switches and eliminating the need for external high-voltage digital control signals.

IV. Conclusion

This brief presents a CCDD-based reconfigurable rectifier designed for high-efficiency energy harvesting under ultra-low RF energy input conditions. Leveraging the proposed knockout reconfiguration method, the RFEH front-end autonomously reconfigures itself within a limited 1.2 V supply voltage, ensuring a stable 1.3 V to 1.8 V output across a wide range of RF input conditions to support ultra-low-power circuits and achieve self-sustained operation. Fabricated in a 40 nm CMOS process and operating at a 900 MHz RF input, the measured results demonstrate a high efficiency of 42.8% and a high sensitivity of –23.6 dBm. Compared to other reconfigurable rectifiers, this design features lower extra power requirements and a wider stable output range for low-power systems.

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